

Fabian Lander

fabian.lander@mis.mpg.de

Education

PhD Mathematics

2024–Present

Max Planck Institute for Mathematics in the Sciences, Leipzig,
Research Focus: Affine measured foliations and dilation structures
Advisor: JProf. Dr. James Farre

MSc Mathematics

2021–2023

Heidelberg University, GPA: 1.3
Thesis: Vertical Foliations of Triangulable Dilation Tori
Advisor: JProf. Dr. James Farre

BSc Mathematics

2017–2021

Heidelberg University, GPA: 1.8
Thesis: Polygonal Symplectic Billiards
Advisor: Prof. Dr. Peter Albers

Work Experience

Research Assistant (Mathematics Lab, Leipzig)

2024–Present

Developing interactive visualizations for mathematical research and outreach
Collaborating with researchers on visual tools for geometry and dynamics
Supporting public engagement through exhibits and demonstrations

Research Assistant (University of Heidelberg/HEGL)

2022–2024

Developed real-time visualization tools for flat surfaces using WebGL/Three.js
Supervised student projects in geometric computing
Collaborated on outreach events for HEGL Student Seminar

Research Assistant (University of Heidelberg/SFB 191)

2021–2024

Developed high-performance symplectic billiard dynamics simulator (C++/OpenGL)
Co-authored a paper on periodic orbits in symplectic billiards
Implemented discrete dynamical system visualizations

Teaching Assistant in Analysis 1-3 (University of Heidelberg)

2020–2022

Conducted tutorials in Real Analysis and Measure Theory for 300+ students,
delivering four lessons per week.

Publications and Preprints

The two most symmetric flat tori as eight-vertex paper tori, arXiv:2609.10703 (2026). 26 pages. Preprint.

Symplectic Tiling Billiards on Complete Affine Tori, with Charles Daly, arXiv:2608.28894 (2026). 46 pages. Preprint.

Illustrating Hyperbolic Surfaces with Mesh Embeddings, with Erik Löffelholz, Diaaeldin Taha, Steve Trettel and Anna Wienhard, in *Proceedings of Bridges 2026: Mathematics, Art, Music, Architecture, Culture*, pp. 197–204, Bridges Organization, 2026.

Immersive Visualization of Flat Surfaces Using Ray Marching, with Diaaeldin Taha, in *Proceedings of Bridges 2025: Mathematics and the Arts*, pp. 191–198, Bridges Organization, 2025.

Symplectic billiards for pairs of polygons, with Peter Albers and Jannik M. Westermann, arXiv:2402.12244 (2024). 38 pages. Accepted for publication in the *Arnold Mathematical Journal*.

Illustration als mathematische Forschungstechnik, with Tim Reinhardt and Martin Skrodzki, *Mitteilungen der Deutschen Mathematiker-Vereinigung* 34 (2026), no. 3, 153–158.

Selected Projects

Symplectic Billiards Simulator

C++/OpenGL, 2020

High-performance research tool for dynamical systems analysis
github.com/SiegfriedFabian/Billiards

Raymarching Engine

JavaScript/WebGL, 2022

Interactive visualization of locally flat surfaces
fabianlander.github.io/apps/raymarchingflatsurfacesapp

Flat Tori Gallery

Three.js/WebGL, 2026

Interactive isometric embedding experiments of flat tori in $\mathbb{R}^3$ and S^3
fabianlander.github.io/apps/flatTori

Projective Point Cloud Viewer

Three.js/JavaScript, 2025

Visualization of the degenerate coaffine representation from the work of James Farre and Marit Bobb
fabianlander.github.io/apps/pointCloudViewer

Interactive Pseudosphere with Hyperbolic Tiling

Three.js/GLSL, 2026

Real-time (2,3,7) triangle tiling on surfaces of revolution with adjustable ambient curvature, using RK4 ODE integration
fabianlander.github.io/talks/rotationally-symmetric-surfaces